

DARBI AI System Design

AI-Powered Career & Job Matching Platform

After Homepage Modification

System Overview

DARBI is an intelligent platform connecting students with engineering majors, companies with talented graduates, and profes

Key Features

■ Student Portal	■ Company Portal	■ Career Boost
Discover engineering majors	Post job openings	AI career mentorship
AI recommendations	Filter talented students	Skills assessment
View salary data	Connect with graduates	Industry networking

Homepage Design (Updated with 3 Portals)

The updated homepage features three distinct portal options, each with clear visual hierarchy and call-to-action buttons.

Design Elements:

- Dark Navy gradient background (#001a33 to #0a2647)
- Gold (#d4af37) accent borders and highlights
- Large emoji icons (🎓, 🏢, 🚀) for quick visual recognition
- Descriptive text for each portal explaining its purpose
- Clear, contrasting buttons with hover effects

Portal	Icon	Description	Call-to-Action
Student Portal	🎓	Discover engineering majors and get AI-powered career recommendations	Get Started
Company Portal	🏢	Post jobs and find talented engineering students	Sign In
Career Boost	🚀	Advance your career with AI mentorship and industry insights	Explore

Student Portal Dashboard

User Flow:

- 1. Sign Up → 2. Complete Profile (Interests, GPA, Location, Salary Preference)
- 3. Receive AI Recommendations → 4. View Major Details & Courses

Dashboard Components:

- Welcome header with personalized greeting
- 3 Recommended Majors (Biomedical & Chemical Engineering)
- Match score percentage for each major
- Salary progression data (entry, 3-year, 5-year)
- Top job opportunities for each major
- Save major functionality
- Course recommendations specific to major

Component	Details	Color
Welcome Section	Hello [Name]! + Tagline	Dark Navy header
Profile Form	Interests, GPA, Location, Salary	White background
AI Recommendations	3 cards with majors + scores	Gold border accent
Major Card	Title, score, salary, jobs, courses	White + hover effect
Action Buttons	Get Recommendations, Save Major, Back	Gold (#d4af37)

Company Portal Dashboard

User Flow:

1. Login → 2. View Dashboard → 3. Post Job Opening
4. Set Filters (Major, GPA, Skills) → 5. View Matched Students → 6. Contact

Dashboard Components:

- Company profile section
- Posted jobs list with status
- Job creation/posting form
- Student filter panel (by major, GPA, skills)
- Matched student profile cards
- Contact/messaging functionality
- Analytics (applications, saves, views)

Component	Details	Color
Company Header	Company name + industry + contact	Dark Navy
Job Posting Form	Title, Major, GPA, Skills, Description	White form
Active Jobs List	Posted jobs + applications count	White cards
Filter Panel	Major, Min GPA, Required Skills	Light gray
Student Cards	Name, Major, GPA, Skills, Contact btn	Gold borders
Analytics	Total Applications, Saves, Views	Navy backgrounds

Career Boost Dashboard (For Graduates)

User Flow:

- 1. Sign Up → 2. Enter Current Role & Experience → 3. View Dashboard
- 4. Choose Tab (Overview, Mentorship, Skills, Networking)

Dashboard Components:

- Current role & career level display
- AI-assessed career growth percentage
- Next career goal suggestion
- Four main tabs for career development:
 - Overview: Current stats and progress
 - Mentorship: AI career guidance sessions
 - Skills: Assessment and roadmap building
 - Networking: Industry connections & mentors
- Personalized recommendations from Claude AI

Component	Details	Color
Welcome Header	Welcome, [Name] Current Role	Dark Navy bg
Stats Cards	Current Level, Career Growth %, Next Goal	Gold borders
Navigation Tabs	Overview, Mentorship, Skills, Networking	Gold active tab
Mentorship Tab	AI Session button + guidance content	White background
Skills Tab	Skills assessment + growth roadmap	White background
Networking Tab	Industry peers + mentor directory	White background
Action Buttons	Start Session, Take Assessment, Explore	Gold (#d4af37)

Design System

Element	Details
Primary Color	Dark Navy #001a33 - Used for headers, navigation, text
Accent Color	Gold #d4af37 - Used for highlights, borders, buttons
Background	Light Gray #f9fafb - Main background color
Cards	White with shadow - Content containers
Typography	Sans-serif (Arial, Segoe UI) - Clean and modern
Buttons	Gold background, Navy text, 5px rounded corners
Borders	Gold 4px top/side borders on cards for visual accent

Navigation Flow

Homepage → Portal Selection ↓

- Student Portal → Sign Up → Profile Form → AI Recommendations → Major Details
- Company Portal → Login → Job Dashboard → Post Job → View Matched Students
- Career Boost → Sign Up → Career Dashboard → Tab Selection (Overview, Mentorship, Skills, Networking)

All portals have "Back to Home" button for easy navigation.
User authentication via Firebase with secure session management.

Detailed User Flows

Student Portal Flow

1. Homepage: User clicks "Get Started" on Student Portal
2. Sign Up Page: Enters name, email, password, selects student level
3. Firebase Auth: Account created and data stored in Firestore
4. Profile Page: Fills out interests (checkboxes), GPA (input), location, salary preference
5. AI Processing: Claude API analyzes profile and generates recommendations
6. Results: Displays 2-3 recommended majors with:
 - Match score percentage
 - Detailed reasoning
 - Salary progression data
 - Top job opportunities
 - Required skills
7. Actions: Save majors, view courses, back to home

Company Portal Flow

1. Homepage: User clicks "Sign In" on Company Portal
2. Login Page: Enters company email and password
3. Dashboard: Views company profile and list of posted jobs
4. Post Job: Fills form with title, required major, min GPA, skills, description
5. Filter Students: Sets filters to find matching candidates
6. View Profiles: Browse filtered student profiles with details
7. Connect: Message or contact selected students
8. Manage: Track applications and job posting status

Career Boost Portal Flow

1. Homepage: User clicks "Explore" on Career Boost
2. Sign Up: Enters name, email, password, current role, years of experience, major
3. Welcome: Personalized welcome with current role and career level displayed
4. Overview Tab: Shows current position, AI-assessed growth %, next career goal
5. Mentorship Tab: Access AI career mentorship sessions from Claude AI
6. Skills Tab: Take skills assessment and receive development roadmap
7. Networking Tab: Discover industry peers and potential mentors
8. Growth: Continuous AI-powered recommendations based on profile

Technical Implementation

Layer	Technology	Purpose
Frontend	React 18 + Vite	Fast, responsive UI with hot reload
State Mgmt	Zustand	Lightweight state management
Styling	Tailwind CSS	Utility-first CSS framework
Database	Firebase Firestore	Real-time cloud database
Authentication	Firebase Auth	Secure user authentication
AI Engine	Claude API (Anthropic)	AI recommendations & mentorship
Hosting	Netlify	Fast, secure static hosting

Breakpoint	Device	Layout Behavior
< 640px	Mobile	Single column, stacked cards, touch-optimized
640px - 1024px	Tablet	2 columns, full button width, readable text
> 1024px	Desktop	3 columns, full dashboard layout, hover effects

Responsive Design

DARBI is fully responsive and optimized for all device sizes from mobile phones to desktop displays. The Tailwind CSS framework ensures consistent styling and smooth responsive behavior across all breakpoints.

Claude AI Integration

Claude AI (Anthropic) powers three core features in DARBI:

1. Student Major Recommendations:
- Analyzes student profile (interests, GPA, goals)
 - Generates personalized recommendations for Biomedical & Chemical Engineering
 - Provides match scores, salary data, and job opportunities
 - Real-time API calls with structured JSON responses
2. Company-Student Matching:
- Ranks students based on job requirements
 - Filters by major, GPA, skills match
 - Scores candidates for best-fit rankings
3. Career Mentorship (Career Boost):
- AI career coach providing guidance
 - Personalized development roadmaps
 - Industry insights and trend analysis
 - Skill gap identification and recommendations

User Documentation Sections

Section	Content	Audience
Getting Started	System overview, account creation, first steps	All users
Student Guide	Profile setup, interpreting recommendations, saving matches	Students
Company Guide	Posting jobs, filtering students, managing applications	Companies
Career Guide	Mentorship sessions, skills assessment, networking	Graduates
FAQ	Common questions, troubleshooting, support contacts	All users
Technical Docs	API integration, system architecture, data privacy	Developers

Summary

DARBI Phase 2 Update includes:

- 3-Portal Architecture: Student, Company, Career Boost
- Individual Dashboards: Tailored experience for each user type
- AI Integration: Claude API for intelligent recommendations
- Responsive Design: Works on mobile, tablet, desktop
- Firebase Backend: Secure, scalable cloud infrastructure
- Professional Branding: Dark Navy + Gold design system

Design Philosophy:

DARBI prioritizes user experience with clear visual hierarchy, intuitive navigation, and AI-powered personalization. The system serves three distinct user groups with specialized features while maintaining consistent branding and high usability standards.

Next Steps

1. Implement individual dashboards for each portal
2. Create comprehensive user documentation
3. Develop AI mentorship features for Career Boost
4. Add company-student messaging system
5. Implement course and skill tracking
6. Deploy to Netlify for production
7. User testing and refinement
8. Launch and monitor analytics